(3) Long Questions (10 + 20 + 15 + 25 = 70 Points)

Hard Problems of Diffie-Hellman (10 points)

Show that it is possible to solve the Decisional Diffie-Hellman (DDH) problem if one knows how to solve the Discrete Logarithm (DLG) problem. **Show your work!!**.

[replace this with your answer]

Sending a message securely (20 points)

You want to send a secret message M to a law enforcement agency to notify them about some illegal activities you have observed in campus.

You have only access to an insecure medium to communicate with the agency.

You can assume you have not previously communicated with the law enforcement agency. (There are no pre-shared key)

The bad guys are powerful (has a lot of resource) and they know that you are aware of something about their illegal activity but the exact details are unknown to them.

Adversary model:

The bad guys can intercept your communication.

They either want to read your message or want to tamper it on the wire to corrupt your message.

You can assume bad guys cannot stop you from sending the message.

Problem:

Give a detailed protocol with which you can send this message M to the law enforcement officials so that the bad guys cannot read the content of M or cannot evade detection when they mess with your message M.

(Give details regarding any cryptographic mechanisms, blocks, or constructs you use in your protocol. For instance, if you are using keys then figure out how many keys are you using and how you are using cryptographic constructs to securely send the messages. Abstracting/Missing details will cause losing points.)

[replace this with your answer]

Public-key Encryption Scheme's Multiplicative Homomorphism (15 points)

Prove that RSA is a multiplicatively homomorphic encryption scheme. An encryption scheme Enc(.,.) is called multiplicatively homomorphic if and only if for all keys K and message m_1 and m_2, the following holds: Enc(K,m_1) * Enc(K,m_2) = Enc(K, m_1*m_2). (Please consult the extra sheet for a description of the RSA Encryption Scheme).